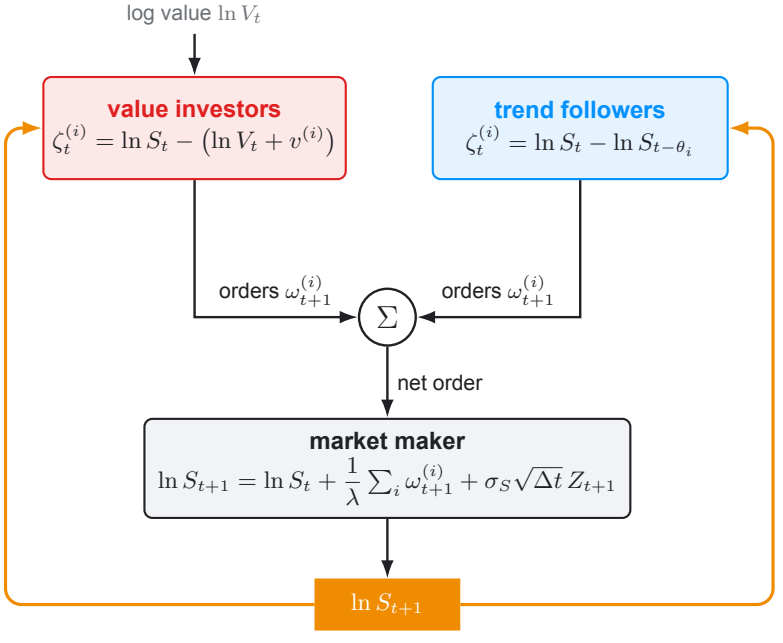
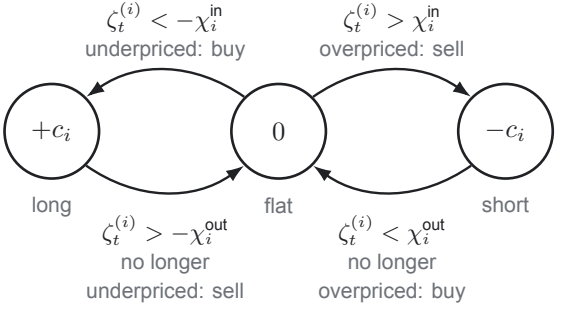
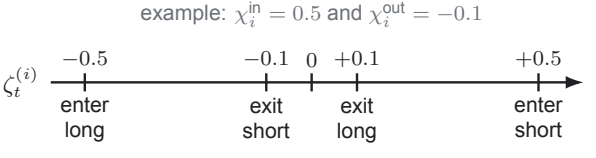

log value $\ln V_t$
value investors
$\zeta_t^{(i)} = \ln S_t - \left(\ln V_t + v^{(i)}\right)$
trend followers
$\zeta_t^{(i)} = \ln S_t - \ln S_{t-\theta_i}$
orders $\omega_{t+1}^{(i)}$
orders $\omega_{t+1}^{(i)}$
Σ
net order
market maker
$\ln S_{t+1} = \ln S_t + \frac{1}{\lambda}\sum_i \omega_{t+1}^{(i)} + \sigma_S\sqrt{\Delta t}\, Z_{t+1}$
$\ln S_{t+1}$
new price, new signals


$\zeta_t^{(i)} < -\chi_i^{\text{in}}$
underpriced: buy
$\zeta_t^{(i)} > \chi_i^{\text{in}}$
overpriced: sell
$+c_i$
0
$-c_i$
long
flat
short
$\zeta_t^{(i)} > -\chi_i^{\text{out}}$
no longer
underpriced: sell
$\zeta_t^{(i)} < \chi_i^{\text{out}}$
no longer
overpriced: buy


example: $\chi_i^{\text{in}} = 0.5$ and $\chi_i^{\text{out}} = -0.1$
$\zeta_t^{(i)}$
-0.5
-0.1
0
$+0.1$
$+0.5$
enter long
exit short
exit long
enter short